Advanced Silviculture

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Silviculture (A34961)

Short Title: Advanced Silviculture
Faculty Science and Computing
Department Science
Cluster Forestry
Author NMCCARTHY
Credits 5 **Level:** Advanced

Description of Module / Aims

This module intends to further explore some of the topics of silviculture that are not generally covered in great detail in the sciences. It aims to elaborate on the new areas of genetics and tree breeding, enabling students to see the potential benefits of tree improvement programmes, and to evaluate the suitability of different approaches to tree breeding for capital intensive and low capital forestry ventures. The module will also introduce the students to tropical and subtropical systems, as well as teaching them about the target seedling concept and advanced nursery practice regimes. The problem case studies of elements of biodiversity and their effect on plantations and their management and development will also be outlined and discussed.

Programmes

`CUTL-0001`	BSc (Hons) in Land Management (in Forestry) (`WD_SBSFO_B`)	1 / 1 / M

Indicative Content

- Introduction, the tropics, plantations; factors affecting plantation development
- Development, organisation and structure of tropical and subtropical plantations
- Forest genetics and tree breeding
- The silviculture and management of veteran trees
- Planting, species selection, cultivation, maintenance, nutrition, thinning, pruning and rotations of tropical plantations
- Introduction to some of the ideas current in the establishment and management of urban woodland and to reassure foresters that they have the knowledge and understanding to work on potentially difficult sites
- Biodiversity and sustainability of plantations
- Advanced forest nursery practices and the target seedling concept

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess the potential benefits of tree breeding and improvement programmes and evaluate the suitability of different approaches to tree breeding for capital intensive and low capital forestry ventures.
2. Assess the constraints of the main silvicultural practices used in the management of forests in the tropics.
3. Critique some of the ideas current in the establishment and management of urban woodland.
4. Choose the appropriate management strategy for veteran trees.
5. Evaluate the target seedling concept and advanced nursery practices.
6. Critique and make informed opinions on articles and papers pertinent to the advanced silviculture.
7. Produce a researched presentation on a topic associated with Advanced Silviculture.

Learning and Teaching Methods

- Lectures.
- Guest seminars.
- Independent learning.
- Case studies.
- Project.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6
Continuous Assessment	50%	
Presentation	30%	7
Case Studies	20%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Independent Learning	87	

Essential Material

- Aldhous, J.R. and W.L. Mason. _Forest Nursery Practice_. UK: HMSO, 1994.
- Dawkins, H.C. and M.S. Phillip. _Tropical moist silviculture and management: A history of success and failure._. UK: CABI, 1998.
- Evans, J. and J.W. Turnball. _Plantation forestry in the tropics_. 3rd ed.. UK: Oxford University Press, 2004.
- Matthews, J.D. _Silvicultural Systems_. UK: Oxford, 1997.
- Miller, R.W., R.J. Hauer and L.P. Werner. _Urban forestry; Planning and managing urban greens_. 3rd ed.. UK: Waveland Press, 2015.

Supplementary Material

- Gunter, S., M. Weber and B. Stimm. _Silviculture in the Tropics (Tropical Forestry)_. UK: Springer, 2011.

Requested Resources

- Lecture Room: Loose Seated